OCTOBER 2026

Open and local AI: no longer a sacrifice

What to decide about models, providers, and AI's physical frontier

Marcelo José Rovai · UNIFEI · AIEng4D





Two paths, one question



BEFORE · UNTIL 2017

In industry

- CEO and VP Latin America at IGT/GTECH
- AT&T, NCR, Dell, and Compaq
- Technology management: P&L, contracts, and vendors



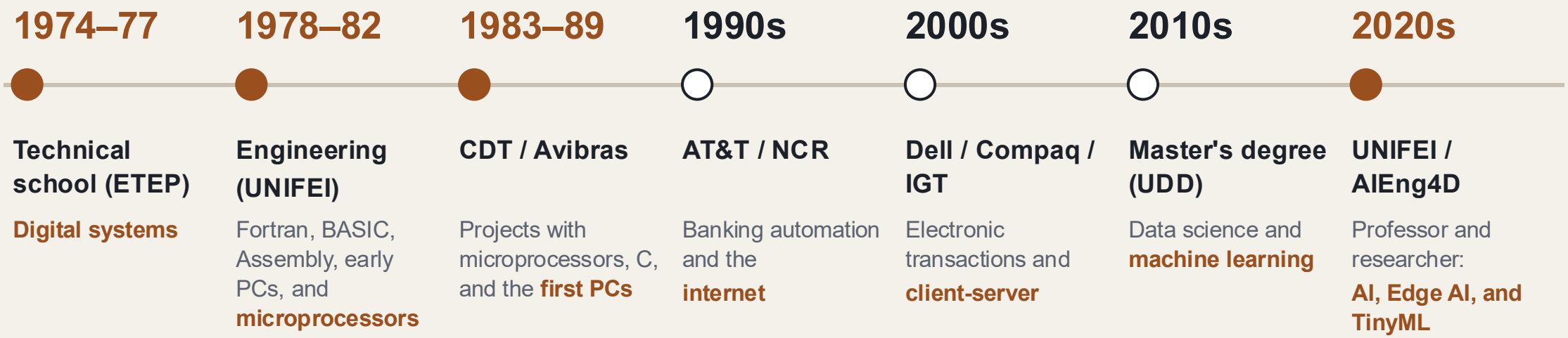
TODAY

In academia

- Professor Honoris Causa, UNIFEI (Brazil)
- Edge AI: AI on low-cost chips, outside the data center
- Co-chair, AIEng4D and EDGE AI Foundation IAP

I didn't run an AI company. I know both sides of the table.

Five decades, seven technology waves

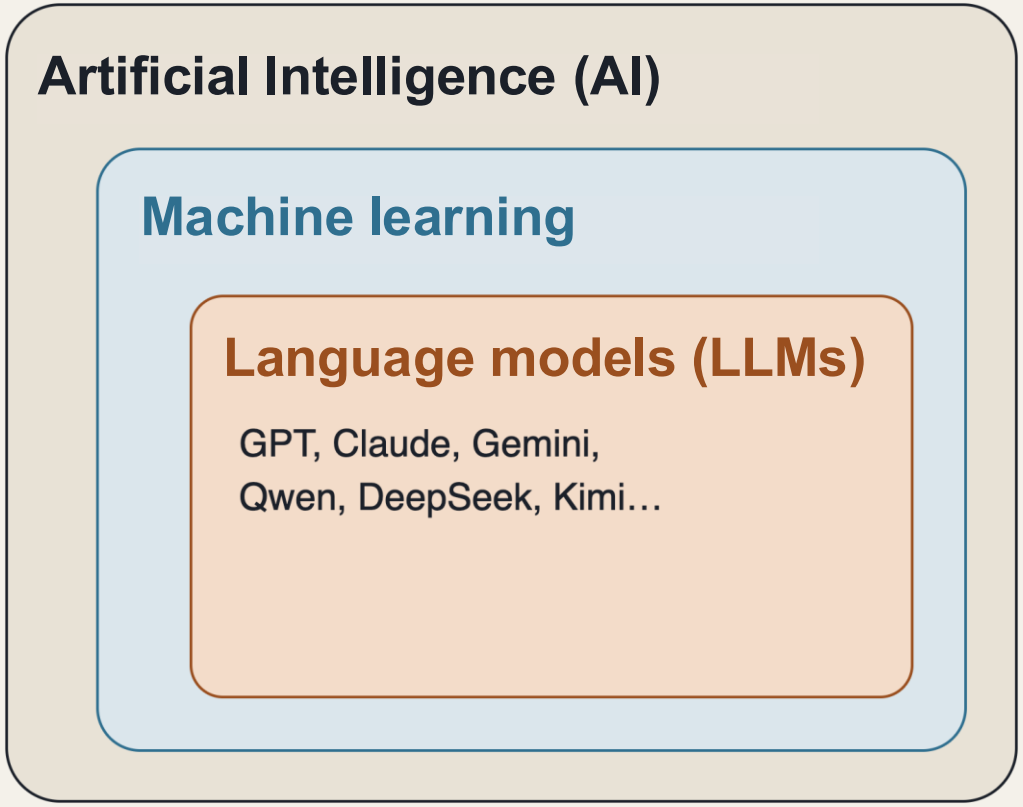


I started by programming microprocessors; today I put AI on microcontrollers. The circle closes.

TODAY'S QUESTION

If your AI provider goes dark on Monday, what stops working in your company on Tuesday?

Three terms, one inside the other



AI

Systems that do tasks we associate with human intelligence.

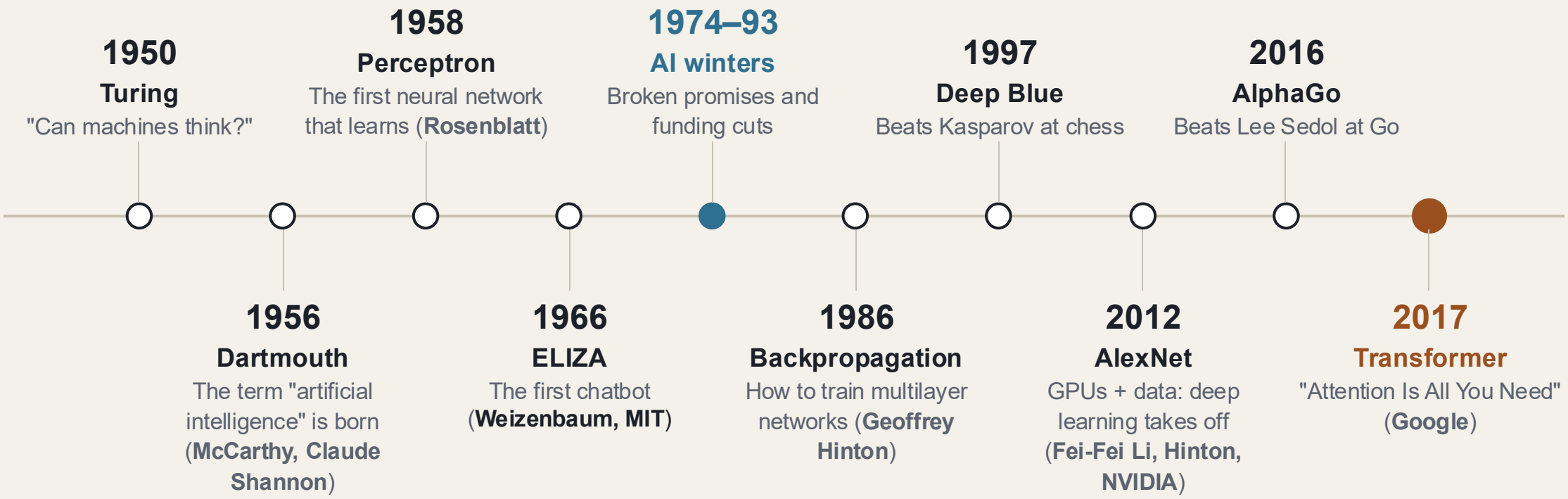
Machine learning

Learns patterns from data. It's already in your company: fraud, demand, maintenance.

Language models (LLMs)

Trained on text at massive scale: they read, write, code, and use tools. What changed since 2022.

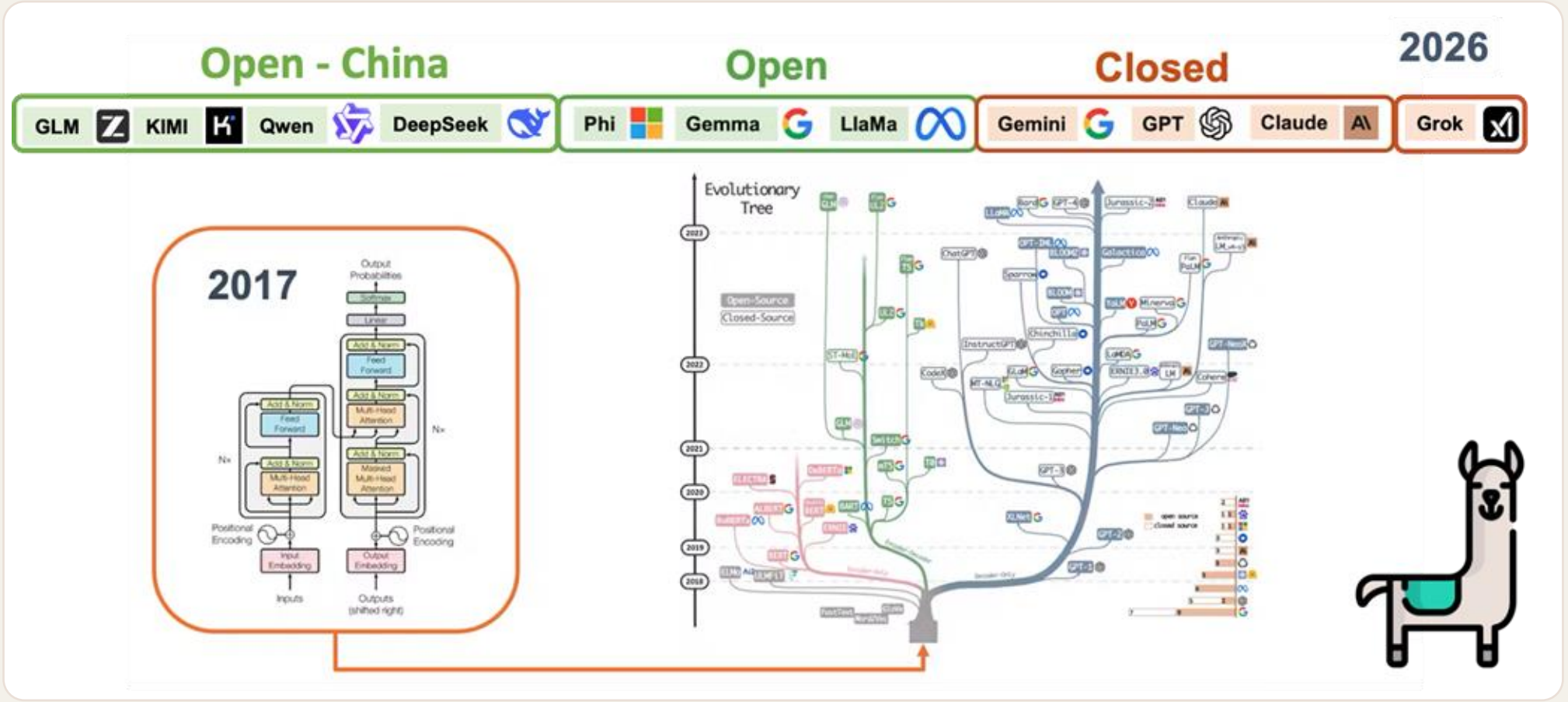
From Turing to the Transformer, in ten milestones



Three ingredients changed everything: data, compute (GPUs), and an architecture that scales.

[History of Artificial Intelligence: A Complete Timeline (1943–2026)]

One architecture, a whole landscape



From the 2017 paper to the 2026 model landscape

Model + harness = agent

Before, assistants: question → answer. Today, agents: goal → loop of actions.

MODEL

The brain

Reasons and generates. Closed (GPT, Claude) or open (Qwen, GLM, Gemma).

+

HARNESS

The body

Tools, memory, permissions, and the decide-and-act loop. (Claude Code, Pilot, Codex, OpenClaw, DSH, Hermes)

=

AGENT

The digital worker

Meets goals: queries systems, reads sensors, and takes actions.

Whoever controls the harness controls the agent. With your own harness, the model becomes a swappable part.

Inside an agent I use every day

DeepSeek Harness (DSH): the model reasons; the harness controls

YOU Ask in plain language · approve risky steps · judge the result

HARNESS · the software around the model

Session context

Which folder it works in
What it may change
Which actions need your approval

The agent loop

Perceive → Plan → Act → Verify

Repeats until the task is done

MODEL

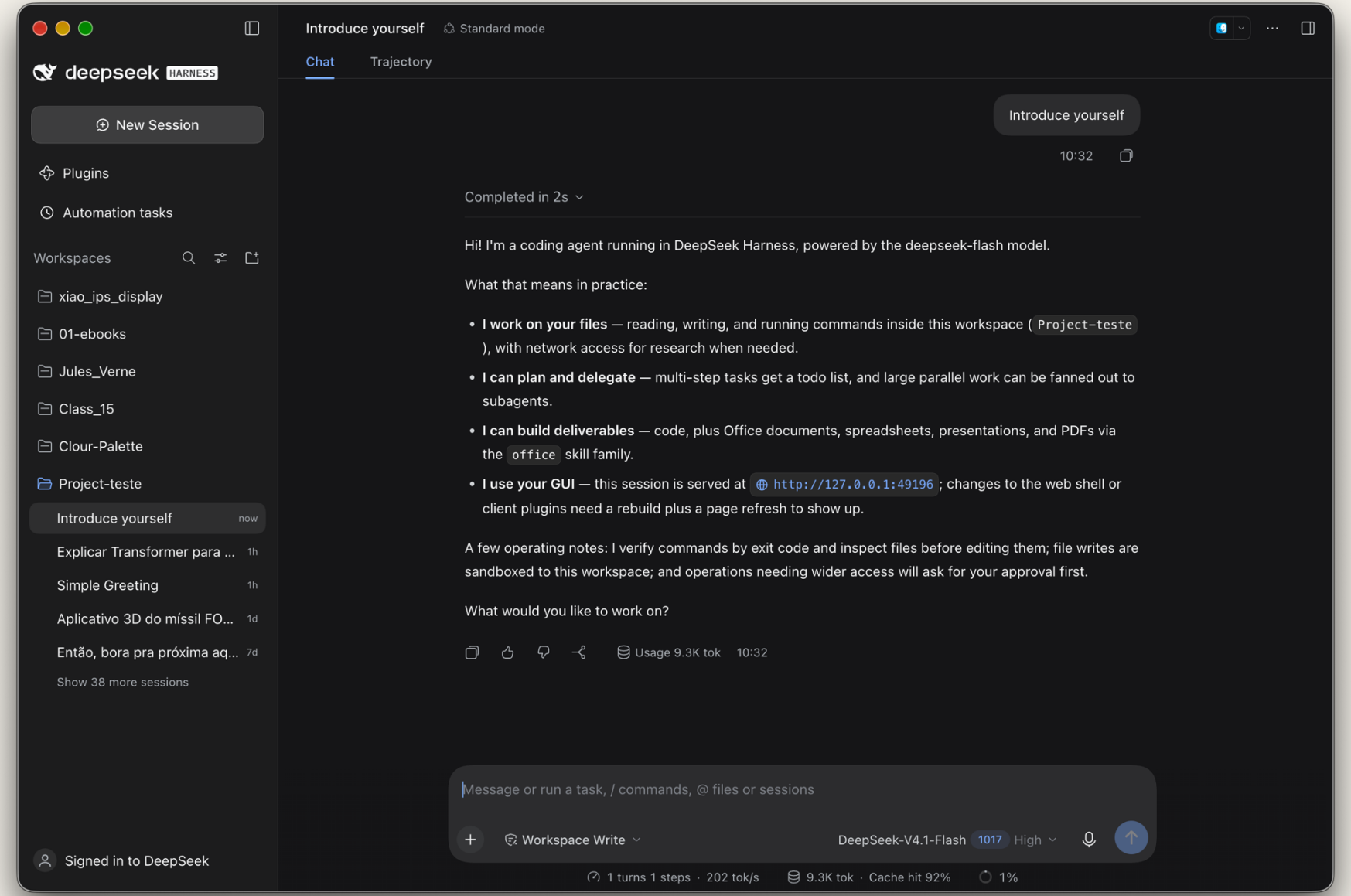
deepseek-flash

Reasons and picks the tools
Never touches a file by itself

Tool executor every action passes the safety rules, and every result is checked

Files | Shell | Search | Web | Subagents | Deliverables

The harness is where a company puts its rules.


deepseek HARNESS
New Session
Plugins
Automation tasks
Workspaces
xiao_ips_display
01-ebooks
Jules_Verne
Class_15
Clour-Palette
Project-teste
Introduce yourself now
Explicar Transformer para ... 1h
Simple Greeting 1h
Aplicativo 3D do míssil FO... 1d
Então, bora pra próxima aq... 7d
Show 38 more sessions
Signed in to DeepSeek
Introduce yourself
Standard mode
Chat
Trajectory
Introduce yourself
10:32
Completed in 2s
Hi! I'm a coding agent running in DeepSeek Harness, powered by the deepseek-flash model.
What that means in practice:
I work on your files — reading, writing, and running commands inside this workspace (Project-teste), with network access for research when needed.
I can plan and delegate — multi-step tasks get a todo list, and large parallel work can be fanned out to subagents.
I can build deliverables — code, plus Office documents, spreadsheets, presentations, and PDFs via the office skill family.
I use your GUI — this session is served at http://127.0.0.1:49196 ; changes to the web shell or client plugins need a rebuild plus a page refresh to show up.
A few operating notes: I verify commands by exit code and inspect files before editing them; file writes are sandboxed to this workspace; and operations needing wider access will ask for your approval first.
What would you like to work on?
Usage 9.3K tok 10:32
Message or run a task, / commands, @ files or sessions
Workspace Write
DeepSeek-V4.1-Flash 1017 High
1 turns 1 steps · 202 tok/s
9.3K tok · Cache hit 92%
1%

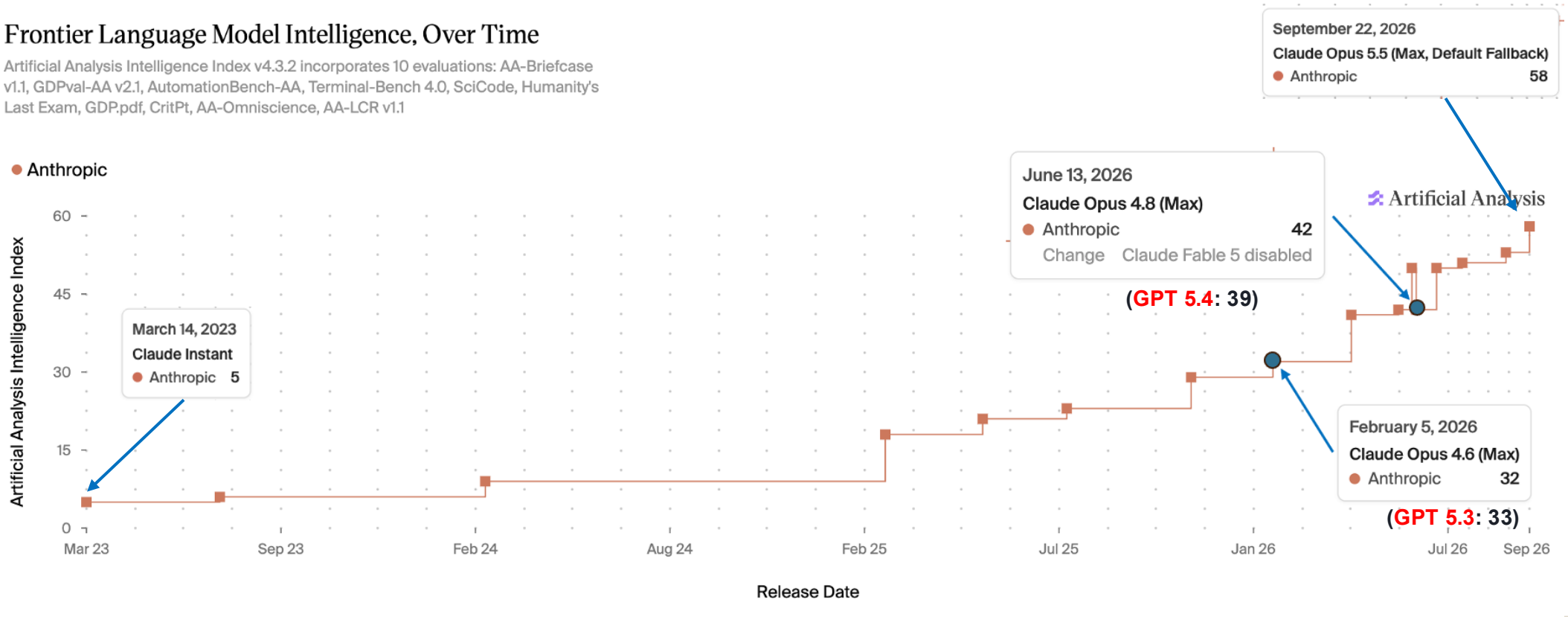
Two independent decisions

	CLOUD · data center	EDGE · Edge AI
CLOSED API only	**GPT, Claude, Gemini** Maximum capability. The vendor controls price, versions, and data.	**Vendor AI** Runs on the device (phone, PC), but it's a black box.
OPEN open weights	**Qwen, DeepSeek, GLM, Kimi, Muse** In your cloud or with the provider you choose.	**Local models** On servers, laptops, cameras, or machines. No internet, no per-token cost.

Open vs. closed: what changes for the company

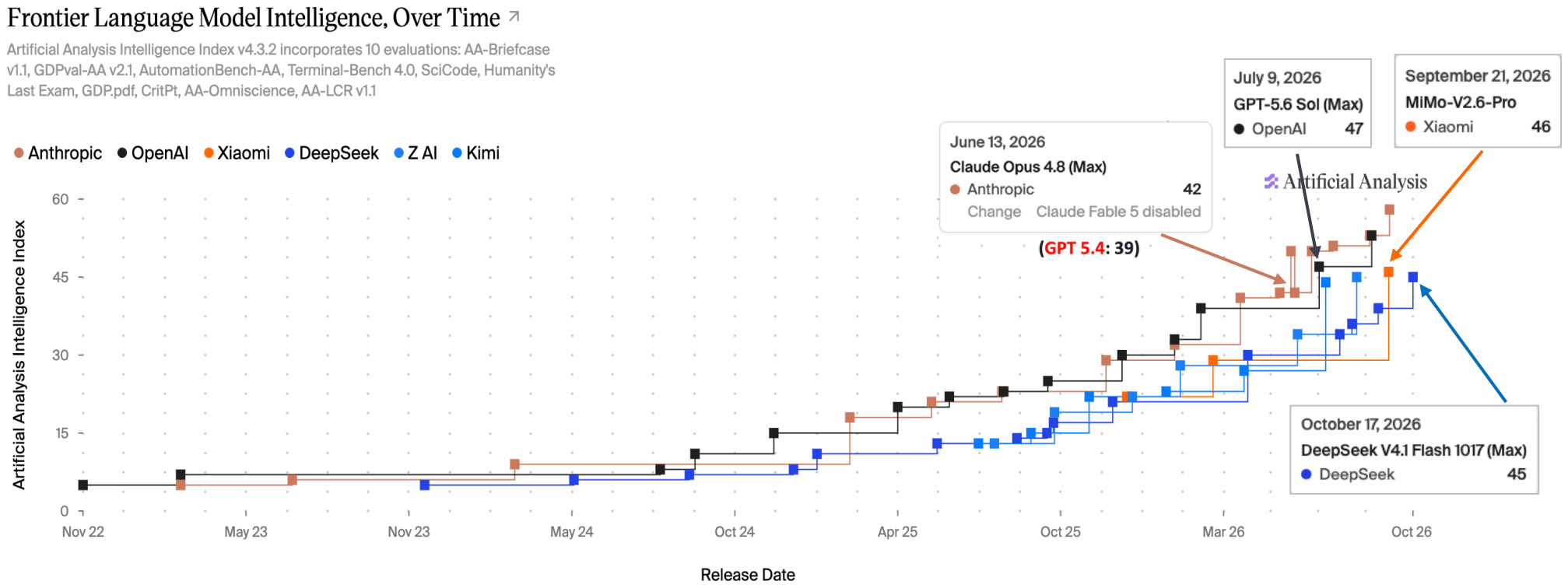
	Closed (API only)	Open (downloadable weights)
Where it runs	The vendor's servers	Wherever you decide
Who changes the model	The vendor: price, version, **retirement**	You freeze the version that works
Your data	Leave the company, under contract	May never leave
Capability	The frontier	A few months behind the frontier
What it requires	Vendor governance	An in-house technical team or an integrator

Anthropic's frontier, month by month



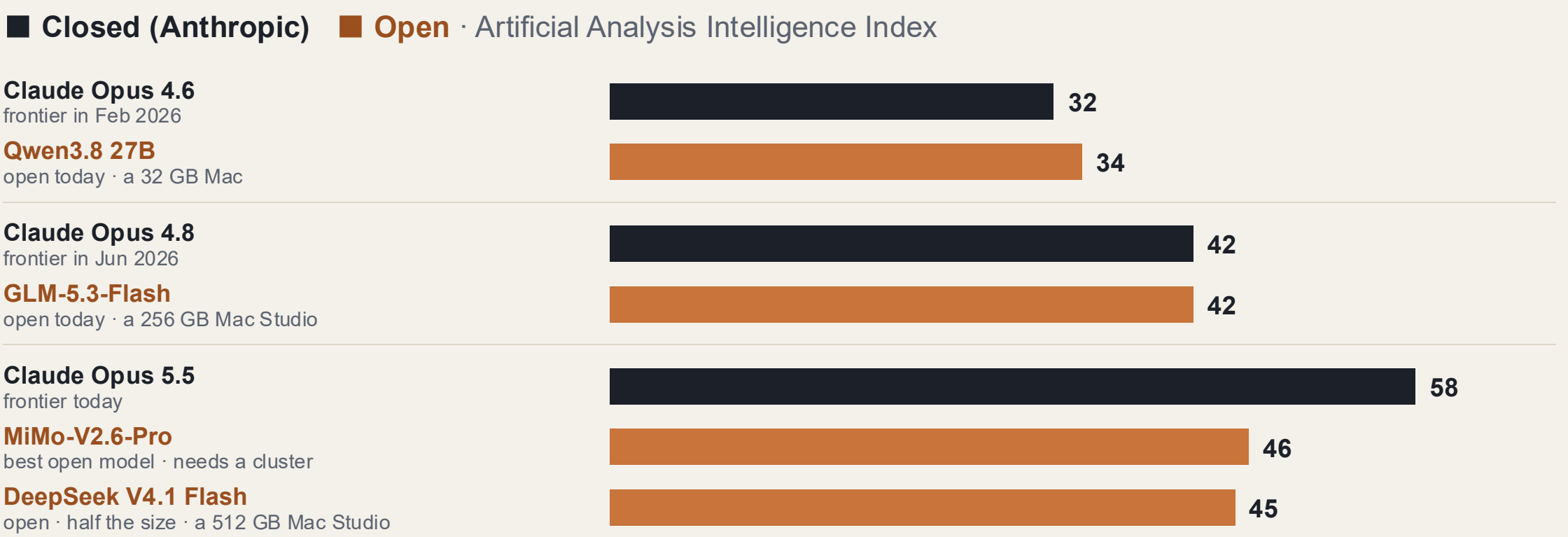
Source: Artificial Analysis, Intelligence Index v4.3.2

Open models catch up with closed ones within months



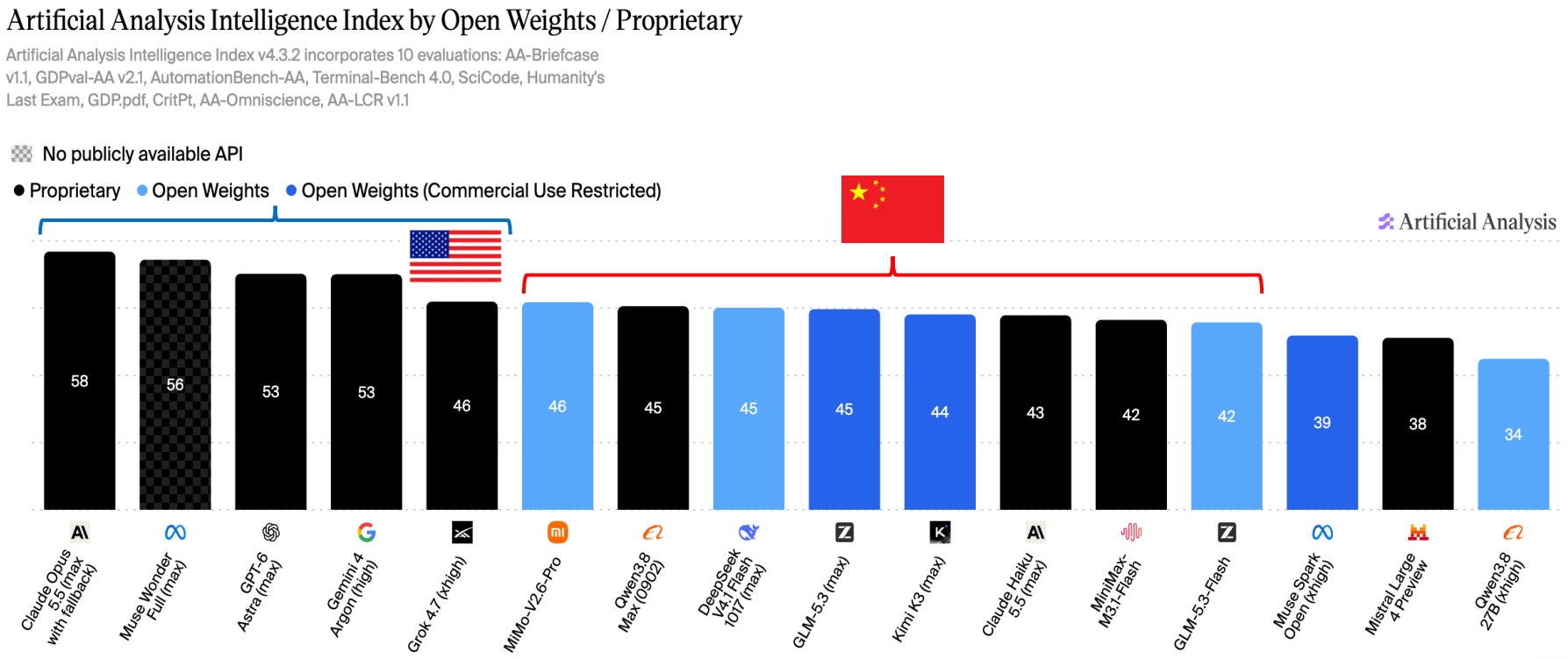
Source: Artificial Analysis, Intelligence Index v4.3.2, Oct 2026 · best model per lab

What was the frontier eight months ago now runs on a Mac



Source: Artificial Analysis Intelligence Index v4.3.2, Oct 2026

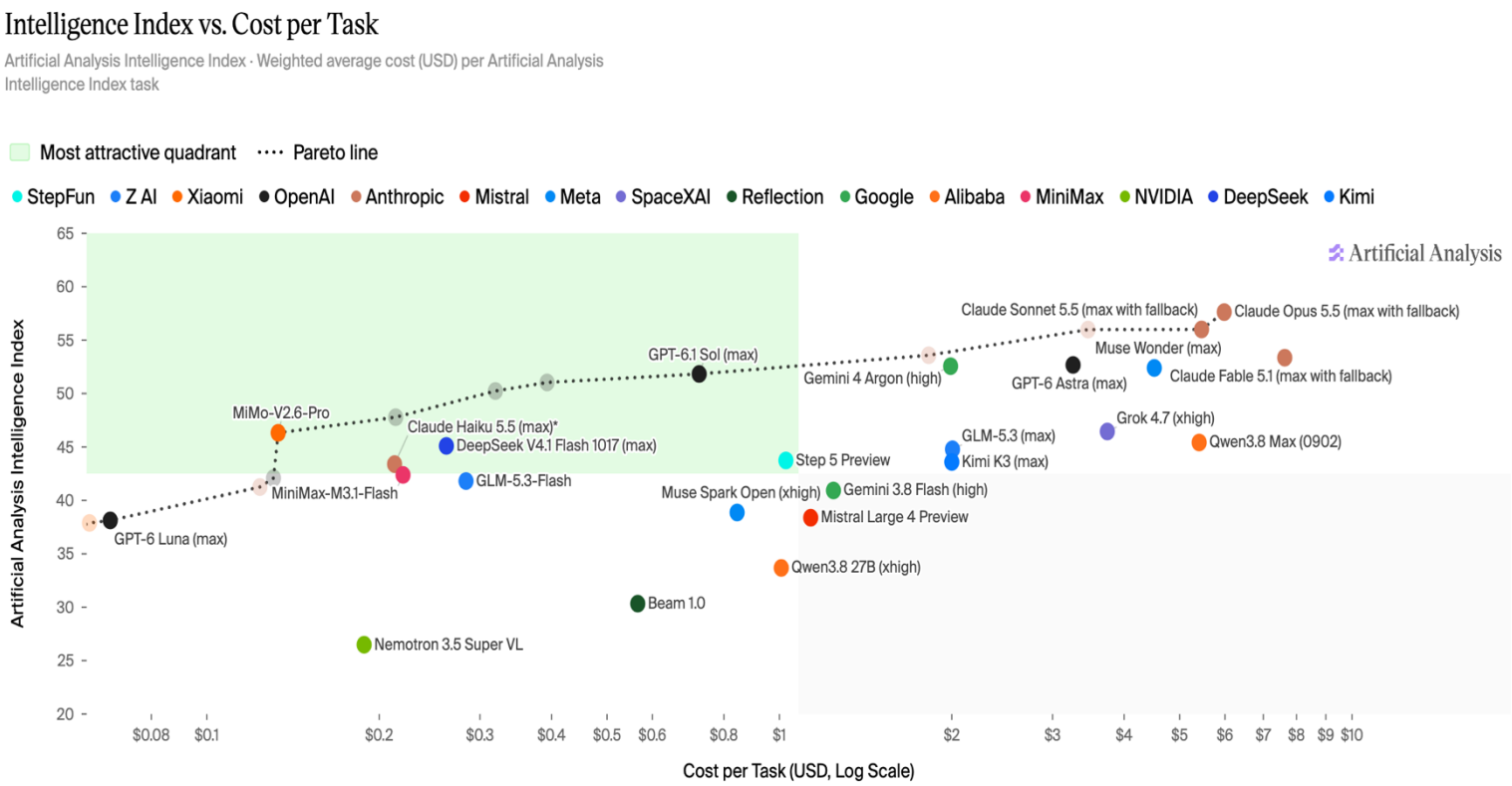
The best closed models are American; the best open ones, Chinese



Source: Artificial Analysis, Intelligence Index v4.3.2, Oct 2026

https://artificialanalysis.ai/

Intelligence vs. cost per task



Source: Artificial Analysis, Intelligence Index v4.3.2 · cost per task via API, log scale

The cost of intelligence is collapsing

280×

cheaper, at equal capability

A GPT-3.5-level model went from US$20 to US$0.07 per million tokens in two years (Nov 2022 – Oct 2024).

Stanford AI Index 2025

15×

cheaper, at equal intelligence

Index 42: GLM-5.3-Flash (open) ≈US$0.28 per task; Claude Opus 4.8 (closed) ≈US$4.10.

Artificial Analysis v4.3.2, Oct 2026 · approx. values

*For the CEO, the question is no longer **"Can we afford AI?"** but:*
"Which task deserves the most expensive model?"

Open doesn't only mean "in-house"

1 · VIA THIRD-PARTY API

Change providers, not models

Several providers serve the same open model: if one raises prices or fails, you switch without rewriting anything.

2 · IN YOUR PRIVATE CLOUD

Data inside the perimeter

Elastic capacity, with the data and the model version under your control.

3 · IN-HOUSE OR EDGE

Full control, no internet

Worth it if data cannot leave, the hardware already exists, or the agent works many hours a day.

Marginal cost of the same coding-agent task (25 turns)

≈US$0.074

Open model (GLM-5.3-Flash) hosted by a third party

≈US$0.003

Your own Mac Studio: electricity only

≈US$0.003

DeepSeek's API with caching, off-peak

Costs: community estimates, Oct 2026 · not independently measured

From the US$79 board to the US$10,799 desktop

US$79

Arduino UNO-Q

An agent that uses tools, with no internet.



~US$1,300

Laptop

A local assistant for documents and code.



~US$2,700

Mac mini 64 GB

Mid-size models for a team.



US$10,799

Mac Studio 256 GB

June's frontier intelligence.



No API key, no usage limits, no per-token charges.

US prices, Aug 2026; outside the US, add taxes and import costs

One model, many agents: a kitchen with one chef

DeepSeek V4.1 Flash on 4× DGX Spark (~US$29,000) vs. DeepSeek's API

1 AGENT

~50–90 tok/s

Feels fast. With a single agent, any serious machine will do.

16 AGENTS AT ONCE

~15 tok/s

Each. That's 247 tok/s in total, shared by all.

DEEPSEEK'S API

~209 tok/s

Each agent gets the full speed. On other providers, between 60 and 220.

~50 s vs. ~3 s

A typical reply: in-house with 16 agents, and on the API

124,000 → 760

Tokens the agent reads to write each reply

98%

Must come from the cache; otherwise each turn takes minutes

The hard part isn't fitting the model: it's serving many agents at the same time.

Sources: community logs and measurements, Oct 2026 · API speed: Artificial Analysis and OpenRouter

When does your own server pay off?

Real case: 111 hours of coding agents in parallel, on a machine with four RTX PRO 6000 cards

THE SAME LOAD, VIA API

US$86

on DeepSeek's API, off-peak: about US$19 a day

≈US$3,500

the same load on Claude Opus 5.5 (my estimate)

Years to pay back the hardware vs. the API



You don't buy it to save money: you buy it for data sovereignty or because a team keeps it busy. Before buying, try the API.

DeepSeek's API, off-peak · hardware at Oct 2026 prices, excluding CPU, memory, and power · Opus 5.5: list prices

The frontier leaders are still losing money

2025 figures, from financial documents reported by Reuters and the Financial Times (not yet audited)

Anthropic

2.76	US$ spent per US$1 of revenue
8.1	US$ billion operating loss
518	US$ billion in compute over 10 years, ~80% fixed

OpenAI

2.60	US$ spent per US$1 of revenue
20.9	US$ billion operating loss
600	US$ billion in compute announced through 2030

It's not a bankruptcy prediction: it's a reason to have a plan B. If the vendor raises prices or changes models, your business shouldn't depend on that decision.

Sources: Anthropic prospectus via Reuters; OpenAI documents via the Financial Times (2026)

The dependency nobody has measured

91% don't fully understand their dependencies on AI vendors, models, and infrastructure

81% would suffer severe or critical disruption if their provider went down for 7 days

71% say switching their main provider or model would be difficult

6 AI disruptions on average over two years, most caused by vendors

IBM IBV and Oxford Economics, "The Calculus of AI Sovereignty," Jun 2026 · 1,000 executives, 16 countries

Which **risks** change with your own open model

Risk	Closed API	Your own open model
Vendor outage	The process stops	Keeps running
Price increase or usage limits	You absorb it	Fixed infrastructure cost
Version retired or changed	The vendor decides	You decide
Sensitive data	Leave under contract	Stay in-house
New risks	—	License, origin, running it yourself

Many of the best open models are Chinese. Run in-house, they send data to no one, but you need to check the license and the biases.

Example: Chile's new personal data regulator

LAW 21.719

Dec 1, 2026

Planned effective date. A bill in the Senate would postpone it to 2027.

FINES

20,000 UTM

Maximum for the most serious violations, enforced by a new Agency.

Every piece of personal data sent to an outside API is processing you'll have to be able to justify.

Check the status of the postponement bill before presenting

Leading companies combine three layers

Orchestrator = software + rules — sends each task to its layer according to company rules

EDGE

Small open model

In sensors, cameras, and machines. No internet, in milliseconds.

ON-PREMISES

Mid-size open model

Confidential data: customers, contracts, and internal documents. Fixed cost.

CLOUD

Open or frontier model

Open via API for high-volume, repetitive work; closed frontier for complex, occasional work.

In practice, the orchestrator is the company's harness.

Example: a distributor with four AI layers

Orchestrator = AI gateway (software) — applies rules defined by the company: IT + legal + business

1 · LOCAL

What's confidential

Contracts, HR, margins, and customer data

Mid-size open model on your own server (e.g., GLM-5.3-Flash)

2 · OPEN, HOSTED

Repetitive, high-volume work

Sort email and complaints, read invoices, describe 20,000 products

Open model via API, with two providers (e.g., Qwen or DeepSeek)

3 · FRONTIER

Complex, occasional work

Market study, due diligence, strategic plan, hard code

Closed model via API (Claude, GPT, Gemini), with no sensitive data

4 · EDGE

Physical, real-time work

Warehouse cameras (inventory, security) and fleet sensors

Small model on the device, no internet

The rule decides where each task goes; people define the rule and review it.

The next wave lives in sensors, cameras, and machines

Mining
Sites with poor connectivity: operator fatigue, predictive maintenance.

Aquaculture (salmon farms)
Cameras at the farm site that count, measure, and alert.

Agriculture and fruit
Quality on the packing line, at belt speed.

Retail
Store analytics without sending customer video to the cloud.

Energy
Monitoring of distributed field assets.

Healthcare
Diagnostic support where clinical data cannot leave.

Five questions to decide

If the answer is yes…	…lean toward
Is the data sensitive or regulated?	Open, on your own infrastructure
Must it work offline or in milliseconds?	Edge (Edge AI)
Is the task complex, low-volume, or exploratory?	Closed, in the cloud
Is it repetitive and high-volume?	Open, in-house or via a provider
Is it a business-critical process?	Two providers or a tested open fallback

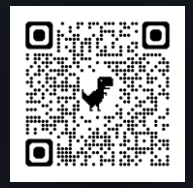
TO CLOSE

A US$79 board runs an agent. A US$10,799 desktop runs the frontier of four months ago. Neither needs anyone's permission.

Questions

https://mjrovai.com/articles/open-and-local-2026/

Marcelo José Rovai



ANNEX A

Large open models: what hardware they need

Model	Total / active	License	Index	Q4 memory	Minimum hardware
MiMo-V2.6-Pro Xiaomi	1.02 T / 42 B	MIT	46	~570 GB	2× Mac Studio 512 GB in a cluster, or 8× H200
GLM-5.3 Z.ai	744 B / 40 B	Restricted	45	~420 GB	Mac Studio 512 GB
Kimi K3 Moonshot	2.8 T / undisclosed	Restricted	44	~1.6 TB	Data center, several GPU nodes
DeepSeek V4.1 Flash **DeepSeek**	552 B (+196 B Engram) / 16 B	MIT	45	~430–510 GB*	Mac Studio 512 GB (1 agent) or 4× DGX Spark
MiniMax-M3 MiniMax	428 B / 23 B	Restricted	29	~240 GB	Mac Studio 512 GB (256 GB very tight)

*DeepSeek already publishes its weights at 4 and 8 bits: 510 GB official, ~430 GB in community builds. The Engram tables can live on an NVMe SSD.

Index: Artificial Analysis v4.3.2 · Memory ≈ 0.56 GB per billion parameters, excluding context cache

ANNEX B

Mid-size and small open models

Model	Total / active	License	Index	Q4 memory	Minimum hardware
GLM-5.3-Flash	321 B / 18 B	MIT	42	~180 GB	Mac Studio 256 GB or 2× DGX Spark
Qwen3.8-Flash-Next	125 B (+51 B n-gram) / 6 B	Restricted	40	~96–114 GB	Mac 128 GB or 1× DGX Spark (tight)
gpt-oss-120b	117 B / 5.1 B	Apache 2.0	12	~63 GB	Mac 96 GB or 1× DGX Spark
Qwen3.8 27B	27 B, dense	Apache 2.0	34	~16–18 GB	Mac 32 GB or RTX 4090/5090
Muse Glimmer (Meta)	~30 B, dense	Apache 2.0	17	~17–20 GB	Mac 32 GB or RTX 4090
Gemma 4 26B A4B	26 B / ~4 B	Open	17*	~15 GB	Laptop with 24–32 GB

Index: Artificial Analysis v4.3.2 · *estimate, independent evaluation pending

ANNEX C

Reference hardware for local models

Hardware	Memory	Bandwidth	US price	Up to which model
RTX 5090	32 GB	1,792 GB/s	~US$4,300–5,000	Qwen3.8 27B, Muse Glimmer
Mac Studio M5 Max	128 GB	614 GB/s	~US$4,800	Qwen3.8-Flash-Next, gpt-oss-120b
1× DGX Spark	128 GB	273 GB/s	US$6,950	gpt-oss-120b, Flash-Next (tight)
2× DGX Spark	256 GB	273 GB/s each	~US$13,900	GLM-5.3-Flash
Mac Studio M5 Ultra	256 GB	1.2 TB/s	US$10,799	GLM-5.3-Flash, at good speed
Mac Studio M5 Ultra	512 GB	1.2 TB/s	Late Oct · n/a	GLM-5.3, DeepSeek V4.1 Flash
4× DGX Spark	512 GB	273 GB/s each	US$27,800 + switch	DeepSeek V4.1 Flash (full)

Prices and specs: Apple and NVIDIA, Aug 2026 (DGX Spark: Oct 2026) · outside the US, add taxes and import costs